

EEBus UC Technical Specification

Optimization of Self Consumption by Heat Pump Compressor Flexibility

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1 Scope of the document

This document describes the Use Case "Optimization of Self Consumption by Heat Pump Compressor Flexibility" (short-name: OHPCF). Chapter 2 specifies the High-Level Use Case. Chapter 3 details the technical solution for SPINE for this Use Case. Within this document, a top-down approach is used to derive the requirements for the technical solution from the High-Level description.

1.1 References

1.1.1 EEBUS documents

[ProtocolSpecification] EEBus_SPINE_TS_ProtocolSpecification.pdf

[ResourceSpecification] EEBus_SPINE_TS_ResourceSpecification.pdf

[SHIP] EEBus_SHIP_TS_Specification_v1.0.1.pdf

1.1.2 Normative references

[RFC2119] IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels
Please see section 1.3.1 for details.

1.2 Terms and definitions

Actor

An Actor models a role within a Use Case definition (e.g. an energy manager or a heat pump).

CEM

Abbreviation for Customer Energy Manager. The CEM is located at the home or premises of the user or in a cloud application. The energy manager enables energy-optimized operation of the connected devices by harmonizing energy demand and availability.

COP

Abbreviation for Coefficient of Performance.

DHW

Abbreviation for Domestic Hot Water.

OHPCF

Optimization of Self Consumption by Heat Pump Compressor Flexibility (short name of this Use Case)

Scenario

Part of a Use Case. Splitting a Use Case into Scenarios helps readers understand the Use Case more quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or optional.

Specialization

Reusable data collection for a specific functionality.

SPINE

Smart Premises Interoperable Neutral-message Exchange: Technical Specification of EEBus Initiative e.V.

1.3 Requirements**1.3.1 Requirements wording**

The following keywords are used:

- SHALL
- SHALL NOT
- SHOULD
- SHOULD NOT
- MAY

Note: They apply only if written in capital letters.

For the meaning of the keywords, please refer to [RFC2119].

1.3.2 Mapping of High-Level requirements

Within the High-Level Use Case description, the following abbreviation is used:

[OHPCF-xyz]

e.g.: [OHPCF-007]

The abbreviation is used to mark High-Level requirements or rules of this Use Case with a unique number xyz. These requirements are referenced throughout the technical solution to show how each High-Level requirement is realized in the technical part.

2 High-Level description

2.1 Introduction

One of the key objectives of a customer energy manager (CEM) is to distribute the electrical power generated by a PV System to the connected electrical consumers while minimizing economic or ecological costs.

One way to achieve this goal is to run an electrical heat pump's compressor preferably at times when enough cheap (or free) PV power is available.

This Use Case describes the combination of a CEM and an electrical heat pump compressor with the goal of optimizing the electrical power consumption of the heating system e.g. according to the available PV power in order to reach economic or ecological goals.

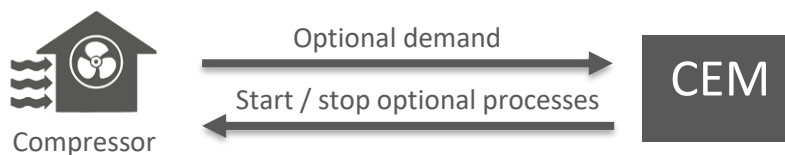


Figure 1: High-Level Use Case functionality overview

Added value: Heat pumps, among others, are energy-intensive devices. To make sure the heat pump is running at the lowest cost, heat pumps must be integrated into the energy management of the smart home.

2.2 User Story as an example

A user wants his energy manager to be capable of managing the demands of the household's devices as best as possible. Therefore, the energy manager activates an optional heat pump compressor's run (e.g. to heat the domestic hot water (DHW) storage) with cheap or even free energy, provided by (e.g.) a local PV system.

2.3 Detailed background information

2.3.1 Overview (informative)

The Use Case is applicable to PV self-consumption systems, where the heat pump compressor can be directly fed by PV energy:

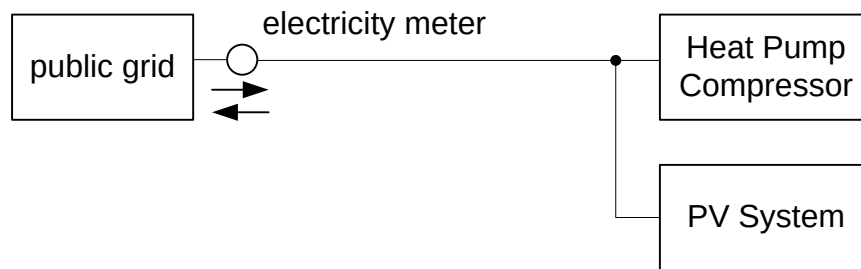


Figure 2: Single electricity meter

The Use Case is NOT feasible for PV systems that feed 100% of produced PV energy into the public grid:

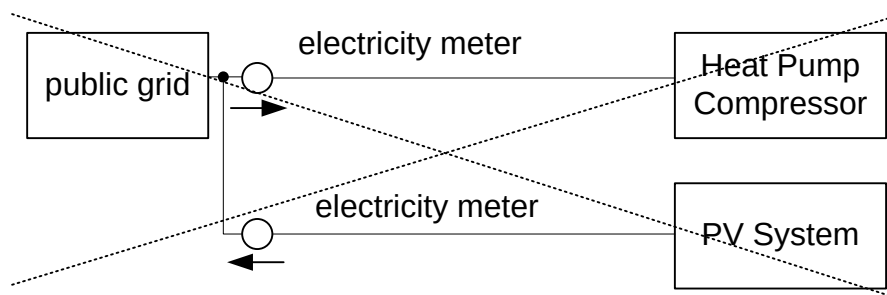


Figure 3: Multiple electricity meters

The PV system described here is a system consisting of one or more PV fields containing several PV modules as well as one or more PV inverters that transform the produced DC power to AC power for subsequent use.

The heat pump system described here is a system consisting of a single electrically driven heat pump compressor as well as thermal storages. These can be domestic hot water storages, heating buffer storages, but also the thermal mass of a building heated or cooled by the heat pump. Heat pump systems are used to serve the thermal demands of a customer's home or premises for domestic hot water, heating, and cooling.

Electrical heat pumps usually contain a compressor as well as an electrical backup heater. The heat pump compressor is the preferred electrical consumer because it is usually able to generate about three to five parts thermal energy from one part electrical energy and typically two to four parts environmental energy (coefficient of performance (COP) approximately 3-5).

The electrical backup heater is usually only used when the compressor alone cannot provide enough thermal energy to serve the thermal demands of the customer. It is not the preferred electrical consumer because it can only generate one part thermal energy from one part electrical energy (COP = 1).

The generated electrical power in a PV system and the consumed power of connected electrical consumers varies over time. As an example, consider the following electrical power generation and consumption:

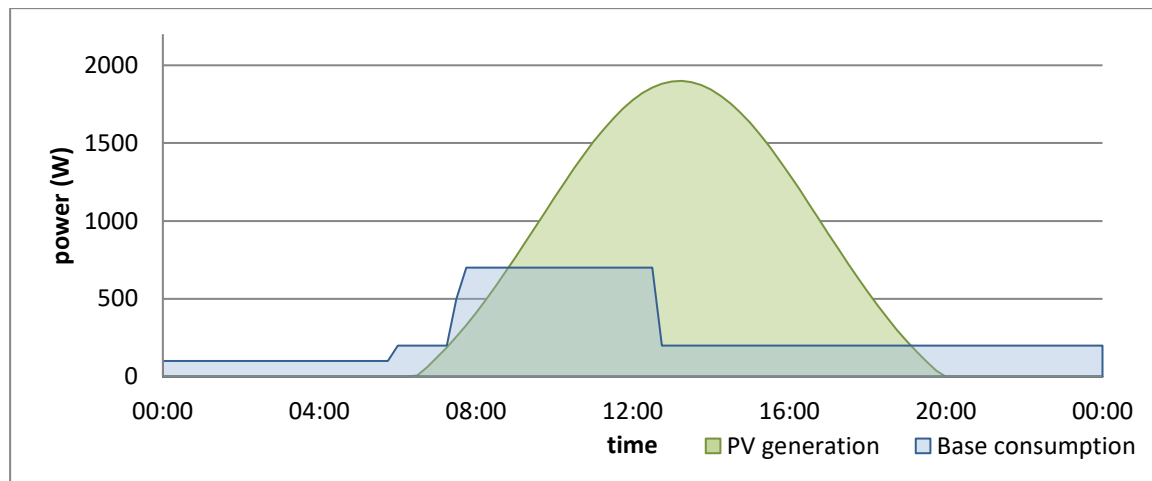


Figure 4: PV generation and consumption

If more electrical power is generated than consumed, the difference between generated power and consumed power is the excess power.

The next diagram shows the resulting electrical excess power of the previous example:

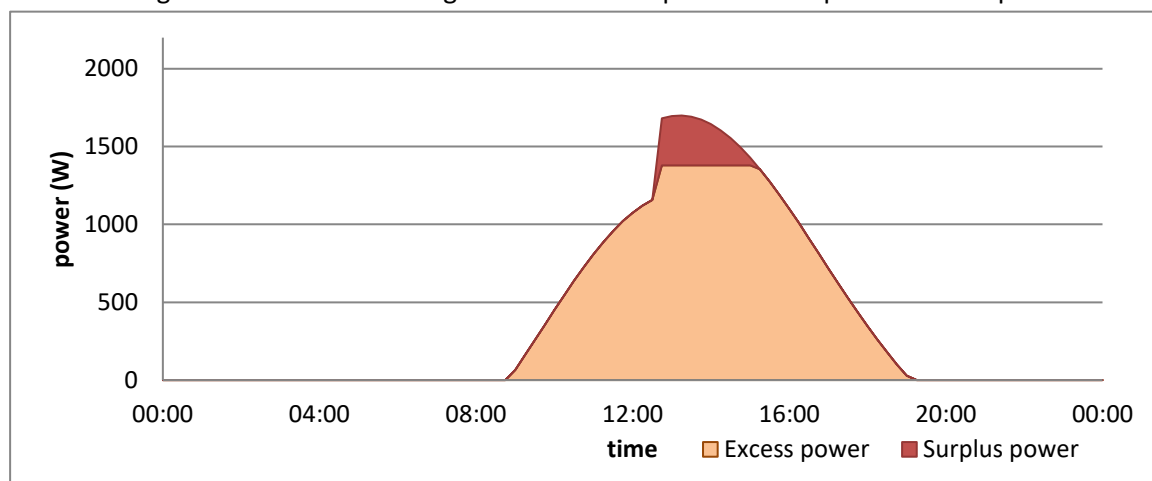


Figure 5: Excess and surplus power

Because the excess power is available over a certain amount of time, the PV system generates excess energy.

The CEM has multiple options to use this excess energy:

- Feed the electrical energy into the grid (resulting in either a reimbursement or penalization)
- Store the electrical energy (e.g., in a battery) and use the electrical energy later
- Operate a device (e.g., a heat pump) and store the energy as thermal energy for later use
- Do not use the energy at all (e.g., reduce power production of the PV inverters)

In many countries, grid operators enforce power feed-in limits, e.g. for peak-shaving purposes.

Hence, option (a) is not always applicable. If the excess power exceeds the feed-in limit – this kind of power is called surplus power – the household or premises must consume the surplus power itself.

Otherwise the customer must pay penalty costs to the grid operator, or the PV inverters have to be reduced to lower the electrical power generation.

221 Also, in many countries, the rate for feed-in electricity is a lot lower than the rate for grid
222 consumption energy, so that it is reasonable to substitute grid consumption by PV consumption as
223 much as possible.

224 In both cases (power feed-in limits / price difference between grid feed in rate and grid consumption
225 rate), the Use Case described here is reasonable.

226 Electrical heat pump systems can help the CEM to use no-cost (e.g. surplus power), low-cost (e.g.
227 excess power), or also "eco-friendly" (e.g. solar power) electrical power more intelligently, taking into
228 account the preferences of the customer (e.g. cost-optimization or eco-friendly optimization).

229 Especially driving the heat pump compressor with PV energy with a COP > 1 to generate thermal
230 energy and to thus replace compressor operation from grid energy by self-produced PV energy is in
231 scope of this Use Case.

232 Driving an electrical backup heater with a COP = 1 with PV energy with a COP = 1 to replace
233 compressor operation with COP > 1 with grid energy is usually not beneficial with low-cost energy,
234 because the thermal energy can still be produced cheaper by the compressor with grid energy than
235 by the backup heater with low-cost (but not no-cost) PV-energy. Therefore, this combination is NOT
236 in scope of this Use Case.

237 Driving an electrical backup heater with a COP = 1 with no-cost PV energy also only makes sense,
238 when the compressor with COP > 1 is not able to be operated (which can have several reasons like
239 too low environmental source temperature or too high thermal sink temperature).

240 Therefore, this Use Case will only describe the energy managed operation of the heat pump
241 compressor, not the electrical backup heater.

242 Instead of just reducing the PV inverters' power generation (due to feed-in limitation, e.g.), the CEM
243 can recommend starting the operation of an electrical heat pump compressor whenever no-cost
244 electrical power is available.

245 If the feed-in remuneration is lower than the electricity costs of a grid consumption, it might also be
246 beneficial to operate an electrical heat pump compressor with low-cost electrical power instead of
247 feeding the power into the grid.

248 In general, the CEM can operate optional processes of an electrical heat pump compressor whenever
249 the costs of operation are very low, i.e. no-cost electrical power is available. The criteria of costs
250 being "low" can typically be adjusted by the customer.

251 The CEM needs to be informed by the heat pump about appliance-specific data that is useful to be
252 known to the CEM in order to optimize the heat pump properly prior to operation. These data may
253 include the minimal time the heat pump compressor must run until it can be stopped again [OHPCF-
254 008] and the minimal time it needs between having been switched off until it can be started again
255 [OHPCF-009].

256

2.3.2 Workflow of the Scenarios

Sections 2.5.1 and 2.5.2 describe technical details of Scenarios 1 and 2. The workflow primarily distinguishes between a "phase A" and a "phase B", which is as well explained in the Scenario sections. Figure 6 shows the workflow of "phase A":

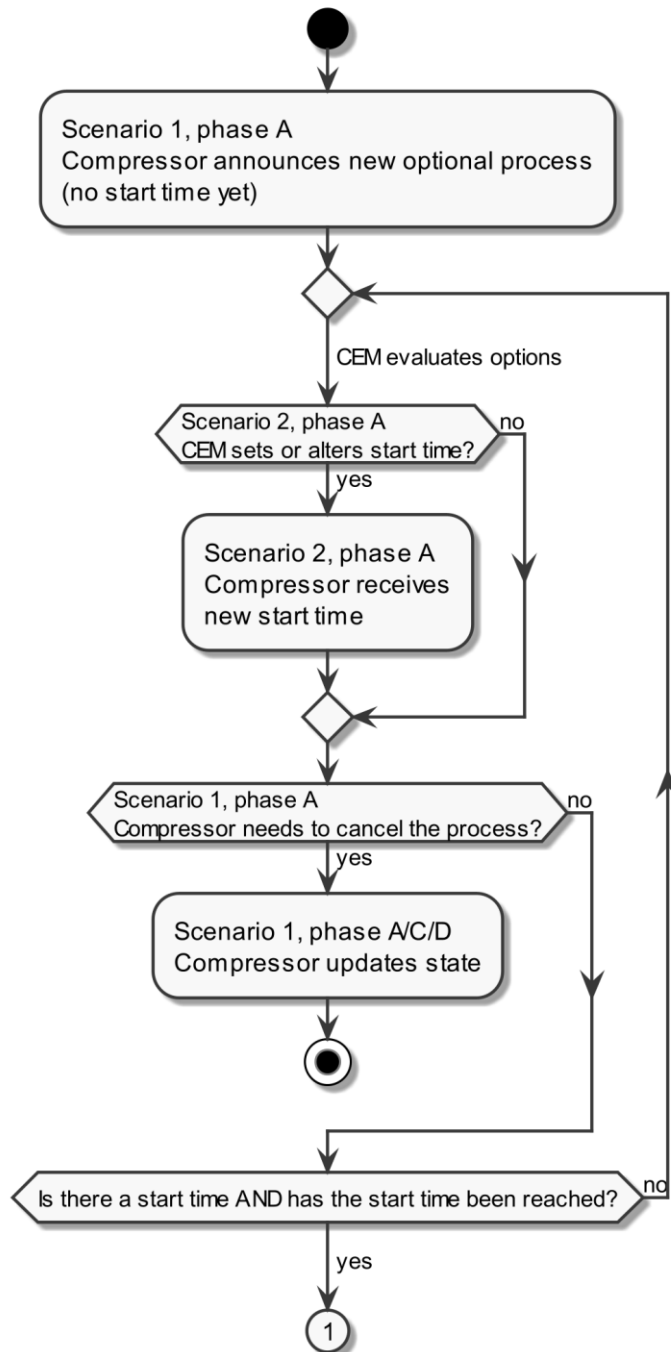


Figure 6: Workflow of "phase A" of Scenarios 1, 2

If the Actor Compressor starts its process as scheduled, the workflow continues with "phase B" at step ① in Figure 7.

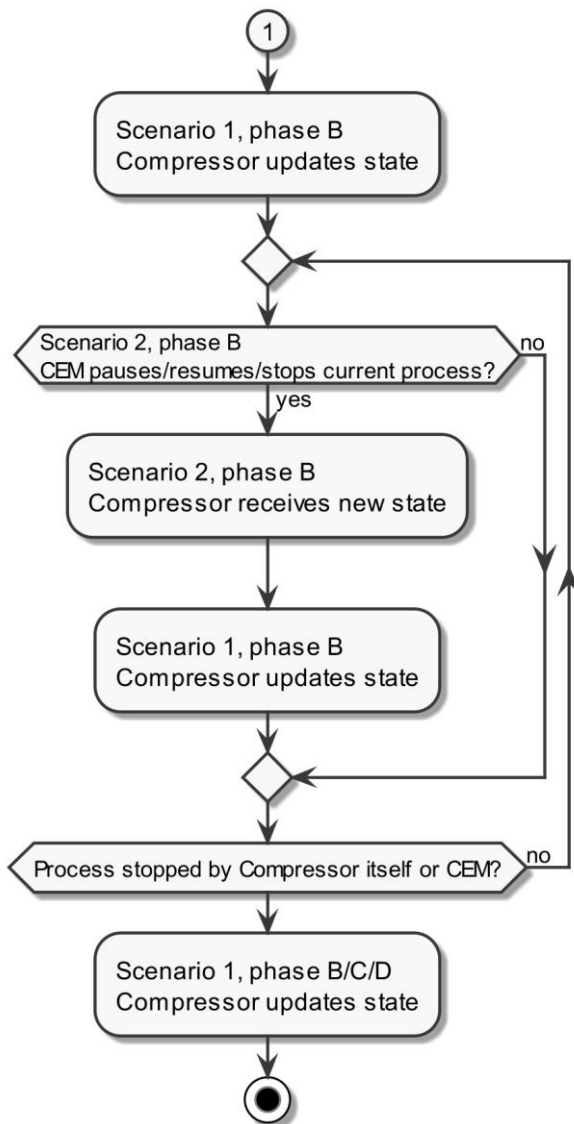


Figure 7: Workflow of "phase B" of Scenarios 1, 2

2.4 Actors

2.4.1 CEM

The Actor CEM represents the energy manager that wants to integrate the heat pump's compressor into the local energy management. In this Use Case, the CEM starts (and stops) the optional heat pump compressor's power consumption (to use the excess or surplus energy provided by a photovoltaic system, e.g.).

2.4.2 Compressor

The Actor Compressor is a component of a heat pump. It consumes electrical energy and generates thermal energy (see section 2.3 for details).

2.5 Scenarios

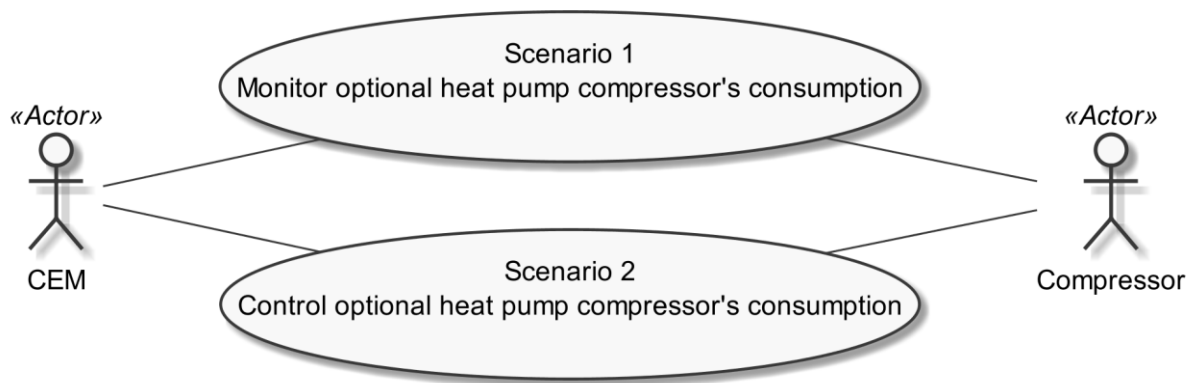


Figure 8: Scenario overview

Scenario number	Scenario name	CEM	Compressor
1	Monitor heat pump compressor's power consumption flexibility	M	M
2	Control heat pump compressor's power consumption flexibility	M	M

Table 1: Scenario implementation requirements for Actors

2.5.1 Scenario 1 - Monitor heat pump compressor's power consumption flexibility

2.5.1.1 Description

The Actor CEM wants to be informed whether the heat pump compressor currently offers an optional consumption of electrical power. If this optional consumption process is activated (i.e. it becomes a scheduled power consumption process) the Actor CEM wants to get the current state of this process. The details on these phases are explained below:

A) Announcement of an optional power consumption process [OHPCF-001]:

This phase applies if the Actor Compressor can offer an optional power consumption. The Actor Compressor announces these data [OHPCF-011]:

1. The availability of an optional consumption of power [OHPCF-011/1].
2. The power value [OHPCF-011/2/1] is denoted as "good approximation" [OHPCF-011/2/2] or maximum power [OHPCF-011/2/3].
3. The consumption will not be started autonomously by the Actor Compressor [OHPCF-011/3].
4. The duration of the power consumption is unknown [OHPCF-011/4] (*1).
5. An indication whether the consumption may be stopped by the CEM [OHPCF-011/5] (*2).

6. An indication whether the consumption may be paused and resumed by the CEM [OHPCF-011/6] (once this optional process has started) (*2).

(*1) Note: A prediction of the maximum energy consumption can typically not be provided by heat pump compressors for these kinds of processes.

(*2) Note: The Actor Compressor SHALL offer at least one option for an interrupt by the Actor CEM [OHPCF-011/7].

This announcement indicates that the Actor CEM may take a proper choice to schedule this process (this is described in Scenario 2).

B) Report of the current state of the scheduled or active power consumption process [OHPCF-002]:

This phase applies after the Actor CEM has taken a proper choice (as described above) and the Actor Compressor accepted this choice. The Actor Compressor announces these data [OHPCF-012]:

1. The start time of the consumption of power [OHPCF-012/1]. The Actor Compressor begins with the power consumption autonomously at this time (provided the consumption is not interrupted) [OHPCF-012/4]. This also means that the Actor Compressor switches autonomously to the respective state (see next item, [OHPCF-012/2/2]).

2. The current state of this power consumption process [OHPCF-012/2]: The process might be scheduled but not started yet [OHPCF-012/2/1], or it might currently consume power [OHPCF-012/2/2], or it might be paused [OHPCF-012/2/3], or it might have been stopped/aborted [OHPCF-012/2/4], or it might have been completed [OHPCF-012/2/5].

3. Current options for the Actor CEM to interrupt or resume the consumption [OHPCF-012/3].

As long as the power consumption has not been stopped/aborted or completed, the Actor CEM may take a proper choice to interrupt or resume the consumption (this is described in Scenario 2).

C) Abortion or completion of a power consumption process according to phase A or phase B [OHPCF-006]:

While being in phase A, Actor Compressor may need to abort the announced optional power consumption process, i.e. before it started. While being in phase B, the current power consumption process may be completed or aborted by Actor Compressor itself or stopped/aborted by Actor CEM.

Upon abortion of a process, Actor Compressor SHALL either announce this abortion explicitly ([OHPCF-006/1]) or it SHALL announce the absence of a power consumption process ([OHPCF-006/2]; see phase D).

Upon completion of a process, Actor Compressor SHALL announce this completion explicitly ([OHPCF-006/3]); see phase D).

D) Absence of a power consumption process according to phase A or phase B [OHPCF-003]:

Before announcement of an optional process (phase A) and after termination of a process (i.e. after phase B/C) the Actor Compressor announces that there is no process.

Note: This Scenario just permits the presence of a single optional or scheduled power consumption process for a given time [OHPCF-013]. I.e. the presence of alternative power consumption processes is not permitted.

2.5.1.2 Conditions

Triggering Event:

The Scenario is triggered by connecting the heat pump's compressor to the CEM.

Pre-condition:

The Actor CEM does not know

- whether the Actor Compressor has an optional power consumption ready to be started or not, or
- if there is already a scheduled power consumption.

Post-condition:

The Actor CEM knows whether the Actor Compressor has an optional power consumption ready to be started or not, or it knows the current state of a scheduled power consumption.

2.5.2 Scenario 2 - Control heat pump compressor's power consumption flexibility

2.5.2.1 Description

This Scenario describes if and how an Actor CEM can advise the Actor Compressor to schedule (activate) an optional power consumption process or to interrupt or resume a scheduled power consumption process:

A) Schedule an optional power consumption process [OHPCF-004]:

This phase applies if the Actor Compressor announced an optional power consumption process (see Scenario 1). If the Actor CEM wants to schedule (activate) this optional power consumption process (*1) it sends a proper request [OHPCF-021] to the Actor Compressor with a start time [OHPCF-021/1] for the power consumption process.

(*1) Note: This is typically the case if the CEM has knowledge of cheap or costless energy produced by a local PV system.

Note: A re-schedule of an already scheduled power consumption process is possible as long as the scheduled process did not start. The messages send to the Actor Compressor are the same as for the first schedule command. If the process already started, only commands from item B) may be used to control the process.

B) Interrupt or resume a scheduled power consumption process [OHPCF-005]:

This phase applies if the Actor Compressor reports a scheduled power consumption process (see Scenario 1). Depending on the reported state and indicated interrupt options, the Actor CEM may send a request [OHPCF-022] to

1. stop (abort) the process [OHPCF-022/1], or
2. pause the process [OHPCF-022/2] (if it is currently in a "running" state), or
3. resume the process [OHPCF-022/3] (if it is currently in a "paused" state).

Note: A request to stop or pause a power consumption is typically applicable if surplus power is not high enough anymore.

Note: The Actor Compressor may also stop the power consumption by itself, i.e. without advice from an Actor CEM (see section 2.5.1.1, phase C). This is typically the case as soon as the hot water storage is completely filled with its maximum temperature.

2.5.2.2 Conditions

Triggering Event:

The Actor CEM wants to start an optional power consumption process or interrupt or resume a scheduled power consumption process.

Pre-condition:

The Actor Compressor announced an optional power consumption process or has a scheduled power consumption process.

Post-condition:

The Actor Compressor reports its new state.

2.6 Dependencies to other Use Cases

2.6.1 "Monitoring of Power Consumption"

The Use Case "Monitoring of Power Consumption" (MPC) enables the Actor CEM to monitor the power consumption of the Actor Compressor.

The Actor CEM SHALL support the Use Case "Monitoring of Power Consumption".

The Actor Compressor SHALL support the Use Case "Monitoring of Power Consumption".

The Actor CEM of this Use Case acts as Actor Monitoring Appliance within the Use Case "Monitoring of Power Consumption".

The Actor Compressor of this Use Case acts as Actor Monitored Unit within the Use Case "Monitoring of Power Consumption".

2.7 Assumptions and Prerequisites

- The electrical heat pump compressor can consume an optional amount of electrical power.
- It is possible that the CEM does not allocate any electrical power to the electrical heat pump compressor. Hence, the correct operation of the heat pump compressor SHALL NOT rely on such optional electrical power.
- Both PV system and heat pump system are connected to the public electricity grid, i.e. the PV system is always able to feed its produced power to the public grid (as far as the grid operator allows) and the heat pump system is always able to obtain its electrical consumption from the public grid if there is not enough PV power available.
- Both PV system and heat pump are connected to the same local electrical grid, i.e. electrical power produced by the PV system can directly be consumed by the heat pump compressor without passing electrical meters that are used to calculate public grid feed-in and consumption at different rates.

3 Technical SPINE solution

3.1 General rules and information

3.1.1 Underlying technology documents

This technical solution relies on the SPINE Resources Specification version 1.1.0 [ResourceSpecification].

For interoperable connectivity this technical solution relies on:

- SPINE Protocol Specification version 1.1.0 [ProtocolSpecification] as application protocol.
- SHIP Specification version 1.0.1 [SHIP] as transport protocol.

3.1.2 Use Case discovery rules

Use Case discovery SHOULD be supported by each Actor. If Use Case discovery is supported the following rules SHALL apply:

- The string content for the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseName" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be "optimizationOfSelfConsumptionByHeatPumpCompressorFlexibility". The string content SHALL only be defined by this Use Case (regardless of the Use Case version).
- The string content of the Element "nodeManagementUseCaseData. useCaseInformation. actor" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be set to the according value stated within the corresponding Actor's section.
- An Actor A that is implemented to support this Use Case specification SHALL set the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseVersion" within the Use Case discovery (please refer to [ProtocolSpecification]) to "1.0.0".
- If an Actor A supports multiple versions of this Use Case with the same major version number, only the highest one SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports multiple versions of this Use Case with the same major version number as supported by Actor A, the Actor A SHOULD evaluate from these versions of Actor B only the highest version number.
- If an Actor A supports multiple versions of this Use Case with different major version numbers, for each major version number only the highest version number SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports only versions with a major version number not implemented by Actor A, it still might be possible to run the Use Case or parts of the Use Case. Therefore, the Actor A should try to evaluate the Actor B as a valid partner for this Use Case.

3.1.3 Rules for "Content of Specialization..." tables and "Content of Function..." tables

3.1.3.1 General presence indication definitions

Abbreviations for the presence indication of Elements listed in the tables are defined as follows:

Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 2: Presence indication description

An Actor MAY support Elements that are not listed in the tables. However, another Actor MAY ignore these Elements.

The presence indications "M", "R" and "O" are always meant relative to the respective parent Element. I.e. if a parent Element is optional ("O") and a child is mandatory ("M") the child Element can only be present if the parent Element is present as well.

Note: The indications and the aforementioned rules apply for "complete messages" (so-called "full function exchange", please refer to [ProtocolSpecification]). In contrast, the so-called "restricted function exchange" is designed to permit exchange of specific excerpts of data, i.e. fewer Elements than potentially available from the data owner (partially even not all "mandatory" Elements).

3.1.3.2 Presence indications for "Content of Specialization..." tables

This section only defines rules for the client side.

Elements that are marked with "M" SHALL be supported by the client in case of readable as well as writeable data. This Element may be optional on the server side.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be supported by the client. In other words: If the server responds with the according Element, the client SHOULD be able to interpret the according Elements.
- "O" means that the data MAY be supported by the client. In other words: If the server responds with the according Element, the client MAY be able to interpret the according Elements.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be written by the client.
- "O" means that the data MAY be written by the client.
- "F" means that the data SHALL NOT be written by the client.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "reply" message: The client MAY ignore the Element.
- In case of a "write" operation to be created: The client MAY set the Element but SHALL consider that the server may ignore the Element.
- In case of a received "notify" message: The client MAY ignore the Element.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If

the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- M, O

- M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.3 Presence indications for "Content of Function..." tables

This section only defines rules for the server side.

Elements that are marked with "M" SHALL be supported by the server in case of readable as well as writeable data. In case of writeable data (marked with "M \W") the server does not need to set the Element, because the Element is set only by the client.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be provided by the server.
- "O" means that the data MAY be provided by the server.
- "F" means that the data SHALL NOT be provided by the server.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be supported. In other words: If the client writes the Element, the server SHOULD accept those messages and the contained Elements.
- "O" means that the data MAY be supported. In other words: If the client writes the Element, the server MAY accept those messages and the contained Elements.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "read" request: The according Element MAY be set in the reply.
- In case of a received "write" operation: The server MAY ignore the Element.
- In case of a "notify" operation to be created: The server MAY set the Element.

Note: The server will only accept write operations if the result fulfils the server Function requirements (permitted values, e.g.). Write operations on Elements that are not writeable MAY result in an error message.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

In some cases, an Element may have a double presence indication, separated by a colon. Examples:

527 - M, O

528 - M \W, O \W

529 The description (rules) of the Element details in such a case the presence requirement. A typical
530 example is a list-based Element where the "first" (or "last") Element has a different presence
531 requirement than the other list Elements.

532

533 3.1.3.4 Cardinality indications on Elements and list items

534 A cardinality indication on an Element or list item expresses constraints on the number of
535 occurrences of a given Element or data set. In this section we use "X" as representation for such an
536 Element or data set. Furthermore, "a" and "b" represent constraints. The following rules apply for
537 the occurrence of "X" and its content related to a specific Scenario (see note underneath the list):

- 538 1. X
539 No cardinality indication.
- 540 2. X (a..b)
541 This means "X" SHALL occur at least "a" times and at maximum "b" times.
- 542 3. X (a..)
543 This means "X" SHALL occur at least "a" times and MAY occur more than "a" times.
- 544 4. X (..b)
545 This means "X" SHALL occur at maximum "b" times and MAY occur less than "b" times (even
546 zero occurrences are permissive).

547 Note: These rules apply only under consideration of presence indications and with regards to the
548 given Scenario or Function definition for this Use Case.

549 The following table is an example to explain this for two different placements.

Scenario {...}: M/R/O \W\ \C	Element	Value	[High-Level Mapping] Element and value rules
...	...	...	...
2: M \W	xFeatureType. xListData. xData. (1..3)		
2: M \W	xId	<*#{1..}>	PRIMARY IDENTIFIER
2: M \W	timePeriod		...
2: M \W	timePeriod. startTime	<xs:duration>	
2: M \W	xSlot. (1..)		
2: M \W	xSlot. xSlotId		...
2: M \W	xSlot. duration	<xs:duration>	...
...	...	...	...

550 Table 3: Example table for cardinality indications on Elements and list items

551 The field

552 xFeatureType. xListData. xData. (1..3)

553 introduces a data pattern (required Elements and values) for "xData" instances used for Scenario 2.
 554 The field itself specifies that such an "xData" instance SHALL occur at least 1 time and at maximum 3
 555 times within "xListData" of Feature Type "xFeatureType". However, this constraint holds only for
 556 Scenario 2 and only if such "xData" are required. In this case, they are required, as the left field

557 2: M \W

558 denotes that this data set is mandatory for Scenario 2.

559 The field

560 xSlot. (1..)

561 expresses that the Element "xSlot" SHALL occur at least one time within its "xData", but MAY occur
 562 more than one time.

563 For the expression "<*(1..)>" of Element "xId" please see section 3.1.3.6.

564 The remaining fields do not have an explicit cardinality indication.

565 Note: Cardinality expressions are also used within placeholder expressions as defined in section
 566 3.1.3.6. In many cases such placeholder expressions define the number of required or permitted
 567 Elements or list items as they explicitly define how many different values for a given Identifier are
 568 required or permitted for a given Scenario.

569

570 **3.1.3.5 Writability and changeability indication**

571 In the same column where the presence indications are denoted, a mark is used to distinguish
 572 between writeable, changeable or readable Elements:

- 573 - Elements that are marked with "\W" are written by a client and SHALL be writeable at the
 574 server according to their presence indications. The client is not obliged to read the according
 575 data. Received notifications do not need to be evaluated.
- 576 - Elements that are marked with "\C" are changed by a client and SHALL be changeable at the
 577 server according to their presence indications. The client is not obliged to read the according
 578 data. Received notifications do not need to be evaluated.
- 579 - Elements that are marked with "\RW" are read and written by a client and SHALL be
 580 writeable and provided by the server according to their presence indications. Received
 581 notifications SHALL be evaluated according to their presence indications.
- 582 - Elements that are marked with "\RC" are read and changed by a client and SHALL be
 583 changeable and provided by the server according to their presence indications. Received
 584 notifications SHALL be evaluated according to their presence indications.
- 585 - Elements that are not marked are only read by a client and SHALL be provided by the server
 586 according to their presence indications. Received notifications SHALL be evaluated according
 587 to their presence indications.

"Writeable" means that the Element and its value may be written by a client. This includes the possibility to modify (if the Element is already present), create (if the Element is not present yet), and delete the Element. The server SHALL adjust its Function according to the received "write" operation (unless the server cannot accept the "write" operation according to section 3.1.3.3).

"Changeable" means that the Element's value may be changed by a client. If the Element is not present at the resource before, it probably **cannot** be created by the client via the "write" operation. In this case the server MAY decline such a message.

Note: "\W" includes "\C" already.

Note: Depending on the resource a client might need to request a proper binding before the server accepts a "write" operation.

3.1.3.6 "Value" placeholders

3.1.3.6.1 Introduction

Specializations may use placeholders to model relations between different Elements or even list items of different Functions. The main purpose is to declare which Identifier values relate to each other. As a Use Case does not prescribe specific values to be used for a given Identifier, a placeholder like "<x1>" can be used in "Value" columns to express the intended relations.

There are two styles to identify a placeholder that can be referenced:

1. <xM>
2. <xM#S>

where

1. "x" is any alphabetical prefix like "m", "t", "ec", "cc", etc.
2. "M" is a (major) number like "1", "2", "15", etc.
3. "S" is a sub-number like "1", "7", "10", etc.

Examples for the first style are "<ec1>", "<z12>". Examples for the second style are "<p4#2>", "<m22#3>". For a given placeholder, only one of the styles can be used.

In addition, there are also styles for placeholders that do not need to be referenced:

1. <*>
2. <*#S>

The second style is only used with so-called cardinality expressions.

3.1.3.6.2 Uniqueness of placeholders

A given placeholder <xM> or <xM#S> represents the same value throughout a given Use Case specification for a given set of its parent Identifier values. This shall be explained in a brief example:

We assume a list item with PRIMARY IDENTIFIER "pId". It also has a SUB IDENTIFIER "sId" with placeholder "<s1>". Then, each occurrence of "<s1>" represents the same value for a given value of pId. This means that "<s1>" of a list item with pId=1 can differ from "<s1>" of a list item with pId=2. But it also means that "<s1>" represents the same value in all list items with pId=1.

Note: Typically, parent Identifiers like "pId" will also be expressed with a placeholder like "<p5>", e.g. In this case, the uniqueness rule applies for "<p5>" likewise.

Note: The uniqueness also applies for placeholders used as FOREIGN IDENTIFIER.

3.1.3.6.3 Placeholder expressions with cardinalities

For some Identifiers, more than one placeholder is needed. Several notations are used for this purpose, which make use of cardinality expressions. The general notation is as follows:

1. <xM#(a..b)>

This is equivalent to this explicit definition:

At least a and at maximum b placeholders of this list: <xM#1> <xM#2> ... <xM#b>

This means that the implementation of a given Use Case (or Scenario) requires a minimum of "a" distinct values of the respective Identifier. In total, there can be up to "b" distinct values of the respective Identifier.

Additionally, the following notations may occur:

2. <xM#(a..)>

This is equivalent to "<xM#(a..b)>" with "b" equal to infinity.

3. <xM#(..b)>

This is equivalent to "<xM#(a..b)>" with "a" equal to zero.

"<xM#(a..)>" has only a lower bound of "a" distinct values, but no upper bound. "<xM#(..b)>", on the other hand, expresses that the Identifier may not be required at all, but it is permitted to have up to "b" distinct values.

Similarly, the cardinality can be used for placeholders that are not referenced, i.e. <*(a..b)> etc.

Note: The cardinality does NOT express which of the sub-numbers have to be used! I.e., it does NOT mean that the Identifiers <xM#1> ... <xM#a> are always used and just those with larger sub-numbers (<xM#a+1> ... <xM#b>) are optional. If, for instance, a placeholder expression "<xM#(3..5)>" is given, it may well happen that an implementation makes use of <xM#2>, <xM#4>, and <xM#5> (i.e., it does NOT use <xM#1>, <xM#3>). Which sub-numbers are used usually depends on other parts of a Specialization and their references to required placeholders, which is explained in section 3.1.3.6.4.

3.1.3.6.4 References to placeholders and relations

According to the styles for placeholders that can be referenced, an enumeration value "e" can refer to a particular placeholder:

- 658 1. e(-><xM>)
 659 2. e(-><xM#S>)

660 This denotes that "e" is to be used with "<xM>" or "<xM#S>", resp.

661 Example: A Specialization contains the Elements "mld" and "phase". "mld" is an Identifier with
 662 placeholder definition <m2#(1..3)>. "phase" is a string that permits the values "a", "b", and "c" using
 663 this expression:

664 "a"(-><m2#1>)
 665 "b"(-><m2#2>)
 666 "c"(-><m2#3>)

667 This expresses that the enumeration value "a" is to be used with the placeholder <m2#1>, "b" is to
 668 be used with <m2#2> and "c" with <m2#3>.

669 Similarly, a placeholder "yN" can refer to a particular placeholder:

- 670 3. <yN->xM>
 671 4. <yN->xM#S>
 672 5. <yN#T->xM>
 673 6. <yN#T->xM#S>

674 where "T" is a sub-number of "yN".

675 It is also feasible to associate placeholders with cardinalities:

- 676 7. <yN#(a..b)->xM#(a..b)>

677 denotes that <yN#1> is to be used with <xM#1>, <yN#2> is to be used with <xM#2>, etc.

678 Note: In this case, the placeholder expressions of yN and xM must have the same cardinality.

679 In some cases, there is a need to express that multiple list items with similar values are feasible or
 680 required, but only particular combinations of these different data are permitted. The following
 681 example shows that several "fData" list items with different "phase" value are required, but that
 682 these list items may only express either the "phase" value combination { "a", "b", "c" } or the "phase"
 683 value combination { "a", "abc", "neutral" }. The permitted combinations are defined in a note below a
 684 table:

Scenario [{...}]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	F. fListData. fData.		
2: M	zld	<z3#(3..5)>	
2: M	phase	"a"(-><z3#1>)	
		"b"(-><z3#2>)	
		"c"(-><z3#3>)	

	"abc"(-><z3#4>)	
	"neutral"(-><z3#5>)	

Table 4: Content of an example Specialization

Note: One of the following combinations SHALL be used at least: {<z3#1>, <z3#2>, <z3#3>} or {<z3#1>, <z3#4>, <z3#5>}.

3.1.3.7 Rules for content of "Value" column

For a given Scenario, the "Value" column may restrict the permitted content of a Function's Element to one or more particular values. This means that Elements with values deviating from the restriction (i.e. from the permitted values) do not belong to the respective Scenario and need to be considered as if the Element is not set. If more than one particular value is permitted for an Element, the values are in a single line each.

If a presence indication is set for the value (in an additional column before the value), the following rules SHALL be applied:

- "M" means that the value SHALL be supported. This means the value needs to be set at a certain point in time (depending on the value rules) or for a certain Element within a list entry.
- "R" means that the value SHOULD be supported.
- "O" means that the value MAY be supported.

If all possible values of a given mandatory Element are optional or recommended and this Element is used for the purpose of the respective Scenario, one of the values SHALL be set. If all possible values of a given optional or recommended Element are optional or recommended, this Element MAY contain also other values, but then this Element SHALL NOT be considered as part of the respective Scenario.

M, R or O may be combined with the suffix "(event)" to express that a supported value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

If no presence indication is set for the value, the following rules SHALL be applied:

- In case of Elements where the server may set or change an Element on its own (see section 3.1.3.5):
 - o within the tables in the "Server data - Resources" sections:
 - the server SHALL support at least one of the listed values.
 - o within the tables in the "Client data - Specializations" sections:
 - the client SHALL support all listed values.
- In case of Elements that are writable or changeable (see section 3.1.3.5):
 - o within the tables in the "Server data - Resources" sections:
 - the server SHALL support all listed values.
 - o within the tables in the "Client data - Specializations" sections:
 - the client SHALL support at least one of the listed values.

723 Depending on the Element, different values may be used during runtime. If this is the case, those
724 rules are described within the value rules.

725 If a value is placed in parenthesis, the corresponding value is a recommendation. The actual value
726 MAY deviate from this, e.g. "(1024)".

727

728 **3.1.3.8 General information on how to interpret the "Content of Function..." and "Content of** 729 **Specialization..." tables**

730 Within the "Client data - Specializations" sections each Specialization is described in an own sub-
731 section with the name "Specialization "<name of the Specialization>" (e.g. "Specialization
732 "Measurement_GridFeedInEnergy"). It contains only one table that includes all Elements needed for
733 this Specialization. The different Functions are mentioned in a continuous row, highlighted with grey
734 background colour. This row contains the following parts:

735 <Feature Type>. <Function>.[<list entry instance name>.]

736 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
737 could be:

738 DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData.
739 deviceConfigurationKeyValueDescriptionData.

740 In the following rows, only the names of the Elements are stated, without the prefix described above.

741

742 Within the "Server data - Resources" sections each Feature Type is described in an own sub-section
743 with the name "Feature Type "<name of the Feature Type>" (e.g. "Feature Type "Measurement)").
744 It contains sub-sections for each Function named "Function "<name of the Function>" (e.g.
745 "Function "measurementListData"). These sections contain one table with all Elements needed for
746 this resource. The list entries are mentioned in a continuous row, highlighted with grey background
747 colour. This row contains the following parts:

748 <Feature Type>. <Function>.[<list entry instance name>.]

749 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
750 could be:

751 Measurement. measurementDescriptionListData. measurementDescriptionData.

752 In the following rows, only the names of the Elements are stated, without the prefix described above.

753

754 For both kinds of tables, the following applies:

755 - Parent Elements are marked with a dot at the end of the name:

756 <parent Element>.

757 E.g.:

758 value.

- If there are sub-Elements, they are described in own rows with the name of the parent Element as prefix, separated by a dot and a blank space:
 <parent Element>. <sub-Element>
 E.g.:
 value. number

3.1.4 Rules for "Feature Types and Functions..." tables

3.1.4.1 Presence indications for "Feature Types and Functions..." tables

The following presence indications are used:

Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 5: Presence indication of Feature Types and Functions support

If at least one Function of a Feature has the presence indication "M", it is mandatory to support the Feature.

3.1.4.2 Rules for "Possible operations" column

Within the "Feature Types and Functions..." tables the column "Possible operations" state whether the Function is read- or writeable (as defined in the detailed discovery mechanism, see [ProtocolSpecification]).

If the "partial" concept (also called "restricted function exchange") SHALL be supported, the following notation is used (separated for read and write access):

read (M). partial (M)

write (M). partial (M)

If the "partial" concept SHOULD be supported, the following notation is used:

read (M). partial (R)

write (M). partial (R)

If the "partial" concept MAY be supported, the following notation is used:

read (M). partial (O)

write (M). partial (O)

The server can decide whether a notification is submitted complete or partial (as described in [ProtocolSpecification]) if not defined differently within this Use Case Specification.

3.1.5 "Actor ... overview" diagram rules

Within the "Actor [...]" overview diagrams in the "Actors" sub-sections the complete functionality of this Use Case is provided, including optional Scenarios. Which Scenarios are optional can be found in Table 1. The Actor MAY have more functionality implemented than needed for this Use Case.

For the following Actor overview example, a brief description of the graphical symbols will be described.

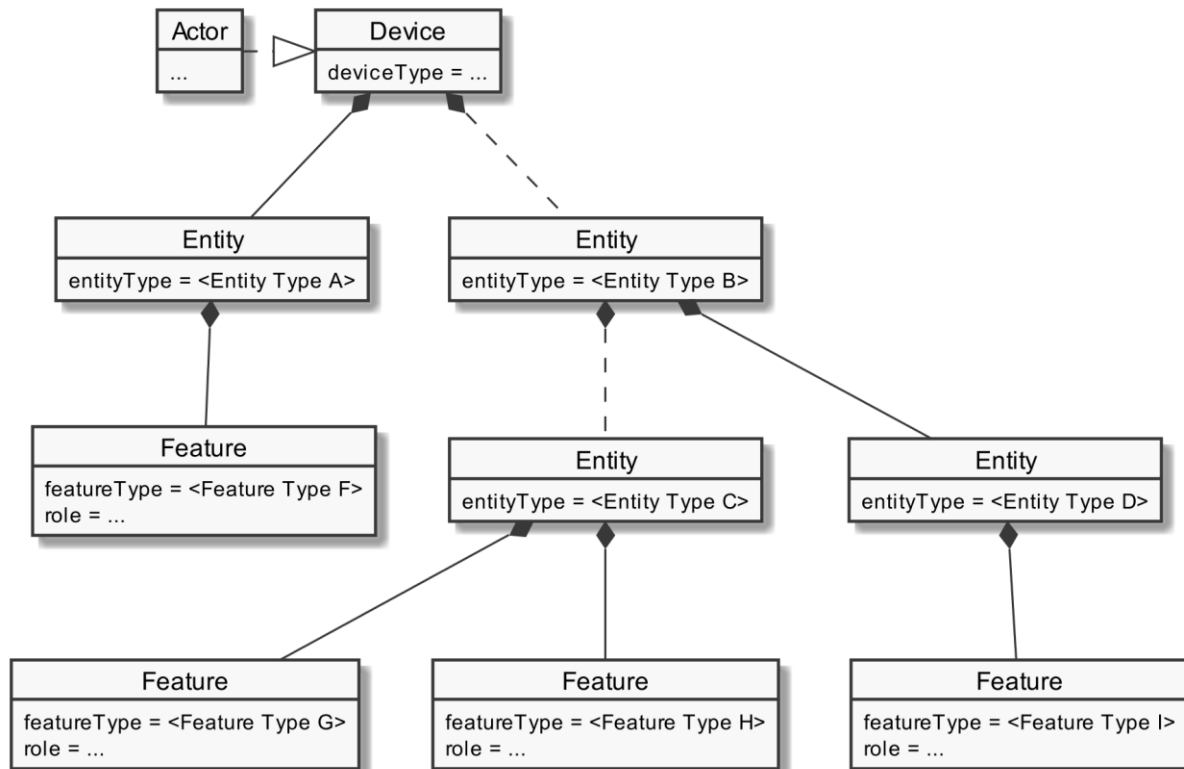


Figure 9: Actor overview example

The solid lines in the figure represent an immediate parent-childhood relation: The Entity with "<Entity Type A>" is a direct child of "Device". The Entity with "<Entity Type D>" is a direct child of the Entity with "<Entity Type B>". All Features are immediate child of the respective Entity.

The dashed lines in the figure express that there MAY be additional Entities between the shown Entities: A vendor's implementation MAY have one or more Entities between "Device" and the Entity with "<Entity Type B>". Likewise, a vendor's implementation MAY have one or more Entities between the Entity with "<Entity Type B>" and the Entity with "<Entity Type C>".

3.1.6 Specializations

Within the "Actors" sub-sections, Specializations are referenced. A Specialization describes a dataset necessary to fulfil the specific requirements of a High-Level Use Case and its Scenarios. Often, data from multiple different Features and Functions are needed to fulfil the requirements. Therefore, a Specialization defines a dataset that may encompass multiple related Functions from one or more different Features.

As different Use Cases sometimes share similar requirements, Specializations are also important from a re-usability perspective. This approach is used to improve consistency across Use Cases and avoid multiple variances of basically the same dataset. This is especially important in the case when an implementation supports multiple Use Cases. E.g. if a power measurement is necessary in two different Use Cases, both Use Cases could define slightly different datasets. In this case, the server as well as the client functionality would have to implement both variances if both Use Cases are supported. This means, depending on the number of Use Cases, two or more datasets need to be generated, transmitted and stored instead of one. Therefore, common Specializations are defined to enable consolidated implementations. Specializations are detailed per Actor, as distinct Actors may use different parts of a Specialization or even have different permissions to change data. In general, this may also depend on the respective Scenario.

If a Feature server can provide the data of a Specialization, the data does not necessarily always need to be available at the Feature server. There might be situations where the user deactivates a Use Case. There may also be other reasons why Use Case data cannot be provided currently. Therefore, a client always needs to be subscribed (as described in section 3.3.4) on the corresponding dataset to stay updated.

The SPINE resource descriptions given in the "SPINE resources of the Actor" sections are derived from the Specializations given in the Actor's overview diagram.

3.1.7 Order of messages within the sequence diagrams

There are several sequence diagrams in this document describing message flows. The order of the messages SHOULD be kept by the communications partners, but there might be cases where a different order makes sense. The communications partners SHALL be able to handle the Scenario functionalities even if the messages are transmitted in a different order by the other Actor(s). The sequence diagrams can be seen as examples.

3.1.8 Further information and rules

3.1.8.1 Frequently used Element rules for the Resource and Specialization tables

This section serves as a collection of rules frequently used by Resource and Specialization tables of the subsequent sections. Each rule applies only where referenced explicitly in the tables.

Note: The purpose of this collection is just to reduce the size of the tables. As such, no rule has a meaning without a reference indicating the required rule. A reference looks like "See [Scaled number rules]", e.g.

[Scaled number rules]:

The sub-Elements "number" and "scale" represent a value according to the following formula:
$$\text{number} * 10^{\text{scale}}$$

3.1.8.2 Rules regarding the usage of time-related information

All time-related information within this Use Case SHALL be presented as relative times. This means absolute times SHALL NOT be used. All start- and end-times SHALL be interpreted as relative to the time the message was transmitted.

3.2 Actors

3.2.1 CEM

3.2.1.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2), this Actor SHALL be denoted as "CEM" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor CEM's resource hierarchy.

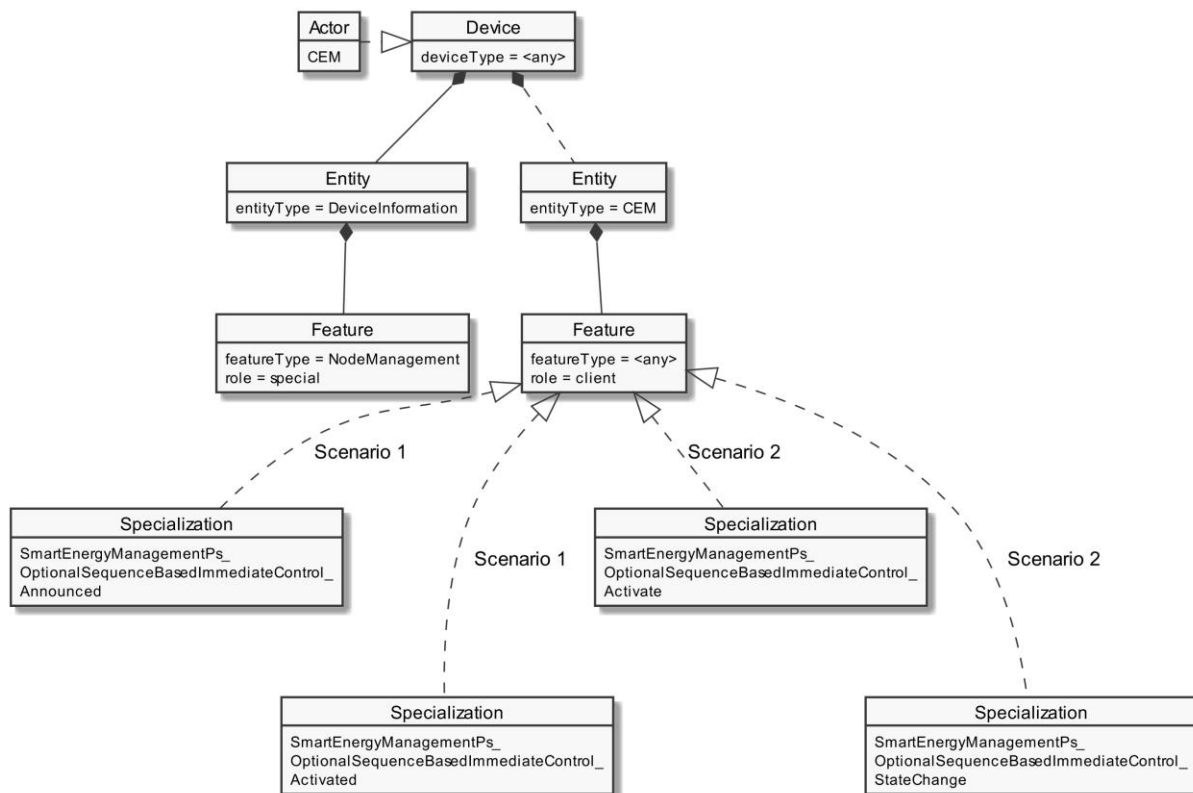


Figure 10: Actor "CEM" overview

The "Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind any entityType. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

871 If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this
 872 data. This means that the Actor accesses the data set described by the Specialization at a
 873 corresponding server Feature. Further details are described in the sub-section "Client data -
 874 Specializations".

875 If a Specialization is connected to a Feature with the role "server", the Actor has the server role for
 876 this data. This means that the Actor must provide the corresponding data set of the Specialization as
 877 part of its Features. Further details are described in the sub-section "Server data - Resources".

878

879 **3.2.1.2 Server data - Resources**

880 As this Actor has only client functionality, no resources are described within this section.

881

882 **3.2.1.3 Client data - Specializations**

883 **3.2.1.3.1 Topic "SmartEnergyManagementPs"**

884 **3.2.1.3.1.1 Specialization**

885 *"SmartEnergyManagementPs_OptionalSequenceBasedImmediateControl_Announced"*

886 This Specialization corresponds to [OHPCF-001] (view of client Actor CEM):

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	SmartEnergyManagementPs. smartEnergyManagementPsData.		
1: M	nodeScheduleInformation.		
1: M	nodeScheduleInformation. nodeRemoteControllable	"true"	
1: M	nodeScheduleInformation. supportsSingleSlotSchedulingOnly		
1: M	nodeScheduleInformation. alternativesCount		
1: M	nodeScheduleInformation. totalSequencesCountMax		
1: M	nodeScheduleInformation. supportsReselection	"false"	
1: M	alternatives. (1..1)		[OHPCF-001], [OHPCF-013]
1: M	alternatives. relation.		
1: M	alternatives. relation. alternativesId	<as1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER of alternatives.
1: M	alternatives. powerSequence. (1..1)		[OHPCF-013]
1: M	alternatives. powerSequence. description.		

1: M	alternatives. powerSequence. description. sequenceId	<s1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER of powerSequence.
1: M	alternatives. powerSequence. description. powerUnit	"W"	If not set, the default value "W" SHALL be assumed.
1: O	alternatives. powerSequence. description. valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
1: F	alternatives. powerSequence. description. repetitionsTotal		
1: M	alternatives. powerSequence. state.		
1: M	alternatives. powerSequence. state. state	"inactive"	[OHPCF-011/1]
1: F	alternatives. powerSequence. state. activeSlotNumber		
1: F	alternatives. powerSequence. state. elapsedSlotTime		
1: F	alternatives. powerSequence. state. remainingSlotTime		
1: M	alternatives. powerSequence. state. sequenceRemoteControllable	"true"	
1: F	alternatives. powerSequence. state. activeRepetitionNumber		
1: F	alternatives. powerSequence. schedule.		[OHPCF-011/3] If Element "schedule" is absent, this denotes that this sequence will not be started autonomously by the server.
1: M	alternatives. powerSequence. scheduleConstraints.		
1: M	alternatives. powerSequence. scheduleConstraints. earliestStartTime		
1: M	alternatives. powerSequence. scheduleConstraints. latestEndTime		
1: M	alternatives. powerSequence. operatingConstraintsInterrupt.		
1: O	alternatives. powerSequence. operatingConstraintsInterrupt. isPausable	"true" "false"	[OHPCF-011/6] If the sequence is pausable by the bound power sequences client, this Element SHALL be present and set to "true". If "isPausable" is NOT set to "true" the Element "isStoppable" SHALL be set to "true" [OHPCF-011/7].

1: O	alternatives. powerSequence. operatingConstraintsInterrupt. isStoppable	"true" "false"	[OHPCF-011/5] If the sequence is stoppable by the bound power sequences client, this element SHALL be present and set to "true". If "isStoppable" is NOT set to "true" the Element "isPausable" SHALL be set to "true" [OHPCF-011/7].
1: O	alternatives. powerSequence. operatingConstraintsDuration.		
1: M	alternatives. powerSequence. operatingConstraintsDuration. activeDurationMin		[OHPCF-008]
1: M	alternatives. powerSequence. operatingConstraintsDuration. pauseDurationMin		[OHPCF-009]
1: M	alternatives. powerSequence. powerTimeSlot. (1..1)		
1: M	alternatives. powerSequence. powerTimeSlot. schedule.		
1: M	alternatives. powerSequence. powerTimeSlot. schedule. slotNumber	<sn1#(1..1)>	SHALL be set as SUB IDENTIFIER.
1: F	alternatives. powerSequence. powerTimeSlot. schedule. defaultDuration		[OHPCF-011/4] If Element "defaultDuration" is absent, the duration of this slot is unknown.
1: M	alternatives. powerSequence. powerTimeSlot. valueList.		
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value (1..2)		If more than one "value" is present, each "value" SHALL have a different valueType.
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value. valueType	"power" [OHPCF- 011/2/2]	
		"powerMax" [OHPCF-011/2/3]	
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value. value.		[OHPCF-011/2/1] [Scaled number rules]
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value. value. number		SHALL be used.
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value. value. scale		SHALL be interpreted. If absent, a default value of "0" applies.

Table 6: Content of Specialization "SmartEnergyManagementPs_OptionalSequenceBasedImmediateControl_Announced" at Actor CEM

890 3.2.1.3.1.2 Specialization

891 "SmartEnergyManagementPs_OptionalSequenceBasedImmediateControl_Activated"

892 This Specialization corresponds to [OHPCF-002] (view of client Actor CEM):

Scenario [{...}]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	SmartEnergyManagementPs. smartEnergyManagementPsData.		
1: M	nodeScheduleInformation.		
1: M	nodeScheduleInformation. nodeRemoteControllable	"true"	
1: M	nodeScheduleInformation. supportsSingleSlotSchedulingOnly		
1: M	nodeScheduleInformation. alternativesCount	1	
1: M	nodeScheduleInformation. totalSequencesCountMax	1	
1: M	nodeScheduleInformation. supportsReselection	"false"	
1: M	alternatives. (1..1)		[OHPCF-002], [OHPCF-013]
1: M	alternatives. relation.		
1: M	alternatives. relation. alternativesId	<as1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER of alternatives.
1: M	alternatives. powerSequence. (1..1)		[OHPCF-013]
1: M	alternatives. powerSequence. description.		
1: M	alternatives. powerSequence. description. sequenceId	<s1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER of powerSequence.
1: M	alternatives. powerSequence. description. powerUnit	"W"	If not set, the default value "W" SHALL be assumed.
1: O	alternatives. powerSequence. description. valueSource	"measuredValue" "calculatedValue" "empiricalValue"	
1: F	alternatives. powerSequence. description. repetitionsTotal		
1: M	alternatives. powerSequence. state.		
1: M	alternatives. powerSequence. state. state	"scheduled" [OHPCF-012/2/1] "running" [OHPCF-012/2/2] "paused" [OHPCF-012/2/3] "completed" [OHPCF-006/3], [OHPCF-012/2/5]	[OHPCF-012/2]

		"invalid" [OHPCF-006/1] [OHPCF-012/2/4]	
1: M	alternatives. powerSequence. state. activeSlotNumber		
1: F	alternatives. powerSequence. state. elapsedSlotTime		
1: F	alternatives. powerSequence. state. remainingSlotTime		
1: M	alternatives. powerSequence. state. sequenceRemoteControllable	"true"	
1: F	alternatives. powerSequence. state. activeRepetitionNumber		
1: M	alternatives. powerSequence. schedule.		
1: M	alternatives. powerSequence. schedule. startTime		[OHPCF-012/1]
1: M	alternatives. powerSequence. scheduleConstraints.		
1: M	alternatives. powerSequence. scheduleConstraints. earliestStartTime		
1: M	alternatives. powerSequence. scheduleConstraints. latestEndTime		
1: M	alternatives. powerSequence. operatingConstraintsInterrupt.		
1: O	alternatives. powerSequence. operatingConstraintsInterrupt. isPausable	"true" "false"	[OHPCF-012/3] If the sequence is pausable by the bound power sequences client, this Element SHALL be present and set to "true". If "isPausable" is NOT set to "true" the Element "isStoppable" SHALL be set to "true" [OHPCF-011/7].
1: O	alternatives. powerSequence. operatingConstraintsInterrupt. isStoppable	"true" "false"	[OHPCF-012/3] If the sequence is stoppable by the bound power sequences client, this element SHALL be present and set to "true". If "isStoppable" is NOT set to "true" the Element "isPausable" SHALL be set to "true" [OHPCF-011/7].
1: O	alternatives. powerSequence. operatingConstraintsDuration.		

1: M	alternatives. powerSequence. operatingConstraintsDuration. activeDurationMin		[OHPCF-008]
1: M	alternatives. powerSequence. operatingConstraintsDuration. pauseDurationMin		[OHPCF-009]
1: M	alternatives. powerSequence. powerTimeSlot (1..1)		
1: M	alternatives. powerSequence. powerTimeSlot. schedule.		
1: M	alternatives. powerSequence. powerTimeSlot. schedule. slotNumber	<sn1#(1..1)>	SHALL be set as SUB IDENTIFIER.
1: F	alternatives. powerSequence. powerTimeSlot. schedule. defaultDuration		[OHPCF-011/4] If Element "defaultDuration" is absent, the duration of this slot is unknown.
1: M	alternatives. powerSequence. powerTimeSlot. valueList.		
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value. (1..2)		If more than one "value is present, each "value" SHALL have a different valueType.
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value. valueType	"power" [OHPCF- 011/2/2]	
		"powerMax" [OHPCF-011/2/3]	
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value. value.		[OHPCF-011/2/1] [Scaled number rules]
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value. value. number		SHALL be used.
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value. value. scale		SHALL be interpreted. If absent, a default value of "0" applies.

Table 7: Content of Specialization "SmartEnergyManagementPs_OptionalSequenceBasedImmediateControl_Activated" at Actor CEM

3.2.1.3.1.3 Specialization

"SmartEnergyManagementPs_OptionalSequenceBasedImmediateControl_Activate"

This Specialization corresponds to [OHPCF-004] (view of client Actor CEM):

Scenario [...]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	SmartEnergyManagementPs. smartEnergyManagementPsData.		

2: M	alternatives. (1..1)		
2: M	alternatives. powerSequence. (1..1)		
2: M	alternatives. powerSequence. description.		
2: M	alternatives. powerSequence. description. sequenceId	<s1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER of powerSequence.
2: M \RW	alternatives. powerSequence. schedule.		
2: M \RW	alternatives. powerSequence. schedule. startTime		[OHPCF-021/1]

Table 8: Content of Specialization "SmartEnergyManagementPs_OptionalSequenceBasedImmediateControl_Activate" at Actor CEM

3.2.1.3.1.4 Specialization

"SmartEnergyManagementPs_OptionalSequenceBasedImmediateControl_StateChange"

This Specialization corresponds to [OHPCF-005] (view of client Actor CEM):

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	SmartEnergyManagementPs. smartEnergyManagementPsData.		
2: M	alternatives. (1..1)		
2: M	alternatives. powerSequence. (1..1)		
2: M	alternatives. powerSequence. description.		
2: M	alternatives. powerSequence. description. sequenceId	<s1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	alternatives. powerSequence. state.		
2: M \RC	alternatives. powerSequence. state. state	"invalid" [OHPCF-022/1]	
		"paused" [OHPCF-022/2]	
		"running" [OHPCF-022/3]	

Table 9: Content of Specialization "SmartEnergyManagementPs_OptionalSequenceBasedImmediateControl_Stop" at Actor CEM

3.2.2 Compressor

3.2.2.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2), this Actor SHALL be denoted as "Compressor" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Compressor's resource hierarchy.

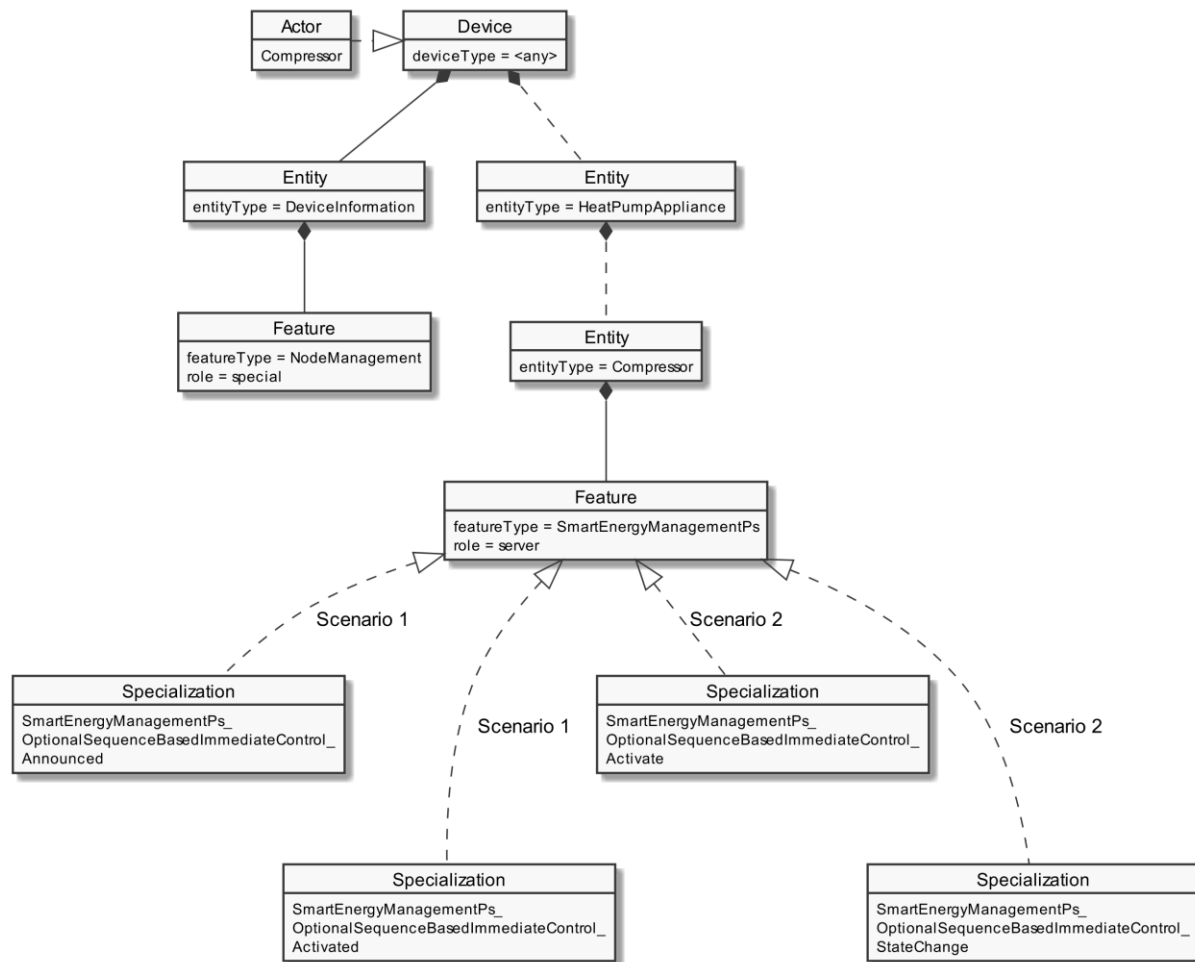


Figure 11: Actor "Compressor" overview

The "Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind the entityType "Compressor", which is a sub-Entity of the Entity "HeatPumpAppliance". The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a

corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

3.2.2.2 Server data - Resources

3.2.2.2.1 Overview

Behind the entityType "Compressor", the Actor Compressor SHALL offer the Feature Types and Functions given in the table below.

Feature Type	Scenario: M/R/O	Function	Possible operations
SmartEnergyManagementPs	1: M 2: M	smartEnergyManagementPsData	[1, 2] read (M). partial (R) [2] write (M). partial (M)

Table 10: Feature Types and Functions used within this Use Case by the Actor Compressor

For each of these Feature Types, the following rule applies: There SHALL be at maximum one Feature with the Feature Type in the Entity.

Note: As a consequence of the previous rule, an implementation may need to have Feature data from different Scenarios/Specializations or even Use Cases in a given Feature.

The Scenario number shows in which Scenarios the Compressor acts as a server and which Feature Types and Functions are relevant in each Scenario.

A detailed definition of the Elements and values that shall be supported in each Function is given in the following sub-sections.

Note: If in the table above "partial" read is not mentioned or is only optional, it still might be mandatory to support partial notifications. The details of "partial" support are described within the Scenario sections.

Note: The presence indications stated above are meant relative to the ones of the according Scenario stated in Table 1. I.e., if a Scenario is optional ("O") and a Feature Type is mandatory ("M"), the Feature Type need only be supported if the Scenario is supported, too.

Note: Further Features MAY be implemented on the same Entities; also, further Functions MAY be implemented in the used Entities.

954 3.2.2.2.2 Feature Type "SmartEnergyManagementPs"

955 3.2.2.2.2.1 Function "smartEnergyManagementPsData"

956 For Scenario 1, the Function corresponds to [OHPCF-011], [OHPCF-012], and [OHPCF-013] (view of
957 server Actor Compressor).

958 For Scenario 2, the Function corresponds to [OHPCF-021] and [OHPCF-022] (view of server Actor
959 Compressor).

960 Note: Several Elements (Element "nodeScheduleInformation" and its child Elements) of this Function
961 SHALL always be set properly even if currently no optional or scheduled power consumption process
962 is available at the Actor Compressor. For more details please see [ResourceSpecification], Feature
963 Type "SmartEnergyManagementPs".

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M 2: M	SmartEnergyManagementPs. smartEnergyManagementPsData.		
1: M	nodeScheduleInformation.		
1: M	nodeScheduleInformation. nodeRemoteControllable	"true"	
1: M	nodeScheduleInformation. supportsSingleSlotScheduling Only		
1: M	nodeScheduleInformation. alternativesCount		
1: M	nodeScheduleInformation. totalSequencesCountMax		
1: M	nodeScheduleInformation. supportsReselection	"false"	
1: M/O 2: M	alternatives. (0..1)		[OHPCF-013] 1: M/O: Mandatory in case of [OHPCF-001], [OHPCF-002], [OHPCF-006/1], [OHPCF- 006/3]. Can be absent in case of [OHPCF-003] or [OHPCF- 006/2].
1: M	alternatives. relation.		
1: M	alternatives. relation. alternativesId	<as1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER of alternatives.
1: M 2: M	alternatives. powerSequence (1..1)		[OHPCF-013]
1: M 2: M	alternatives. powerSequence. description.		
1: M 2: M	alternatives. powerSequence. description. sequenceId	<s1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER of powerSequence.

1: M	alternatives. powerSequence. description. powerUnit	"W"	
1: O	alternatives. powerSequence. description. valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
1: F	alternatives. powerSequence. description. repetitionsTotal		
1: M 2: M	alternatives. powerSequence. state.		
1: M 2: M \RC (event)	alternatives. powerSequence. state. state	"inactive" [OHPCF-011/1]	
		"scheduled" [OHPCF-012/2/1]	
		"running" [OHPCF-012/2/2], [OHPCF-012/4], [OHPCF-022/3]	
		"paused" [OHPCF-012/2/3], [OHPCF-022/2]	
		"completed" [OHPCF-006/3], [OHPCF-012/2/5]	
		"invalid" [OHPCF-006/1] [OHPCF-012/2/4], [OHPCF-022/1]	
1: F (event) 1: M (event)	alternatives. powerSequence. state. activeSlotNumber		If state is set to "running" or "paused" this element SHALL contain the currently active slot number [OHPCF-012]. Otherwise it SHALL be omitted [OHPCF-011].
1: O	alternatives. powerSequence. state. elapsedSlotTime		
1: F	alternatives. powerSequence. state. remainingSlotTime		
1: M	alternatives. powerSequence. state. sequenceRemoteControllable	"true"	
1: F	alternatives. powerSequence. state. activeRepetitionNumber		
1: F (event) 1: M (event) 2: M \RW (event)	alternatives. powerSequence. schedule.		1: F (event): [OHPCF-011/3] 1: M (event): [OHPCF-012/1] 2: M \RW (event): [OHPCF-021/1] If Element "schedule" is absent, this denotes that this sequence will not be started autonomously by the server.

1: M 2: M RW (event)	alternatives. powerSequence. schedule. startTime		1: [OHPCF-012/1] 2: [OHPCF-021/1]
1: M	alternatives. powerSequence. scheduleConstraints.		
1: M	alternatives. powerSequence. scheduleConstraints. earliestStartTime		
1: M	alternatives. powerSequence. scheduleConstraints. latestEndTime		
1: M	alternatives. powerSequence. operatingConstraintsInterrupt.		
1: O	alternatives. powerSequence. operatingConstraintsInterrupt. isPausable	"true" "false"	[OHPCF-011/6], [OHPCF-012/3] If the sequence is pausable by the bound power sequences client, this Element SHALL be present and set to "true". If "isPausable" is NOT set to "true" the Element "isStoppable" SHALL be set to "true" [OHPCF-011/7].
1: O	alternatives. powerSequence. operatingConstraintsInterrupt. isStoppable	"true" "false"	[OHPCF-011/5], [OHPCF-012/3] If the sequence is stoppable by the bound power sequences client, this element SHALL be present and set to "true". If "isStoppable" is NOT set to "true" the Element "isPausable" SHALL be set to "true" [OHPCF-011/7].
1: O	alternatives. powerSequence. operatingConstraintsDuration.		
1: O	alternatives. powerSequence. operatingConstraintsDuration. activeDurationMin		[OHPCF-008]
1: O	alternatives. powerSequence. operatingConstraintsDuration. pauseDurationMin		[OHPCF-009]
1: M	alternatives. powerSequence. powerTimeSlot (1..1)		
1: M	alternatives. powerSequence. powerTimeSlot. schedule.		
1: M	alternatives. powerSequence. powerTimeSlot. schedule. slotNumber	<sn1#(1..1)>	SHALL be set as SUB IDENTIFIER.

1: F	alternatives. powerSequence. powerTimeSlot. schedule. defaultDuration		[OHPCF-011/4] If Element "defaultDuration" is absent, the duration of this slot is unknown.
1: M	alternatives. powerSequence. powerTimeSlot. valueList.		
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value (1..2)		If more than one "value" is present, each "value" SHALL have a different valueType.
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value. valueType	"power" [OHPCF- 011/2/2] "powerMax" [OHPCF-011/2/3]	
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value. value.		[OHPCF-011/2/1] [Scaled number rules]
1: M	alternatives. powerSequence. powerTimeSlot. valueList. value. value. number		SHALL be used.
1: O	alternatives. powerSequence. powerTimeSlot. valueList. value. value. scale		MAY be used. If absent, a default value of "0" applies.

Table 11: Content of Function "smartEnergyManagementPsData" at Actor Compressor

3.2.2.3 Client data - Specializations

As this Actor has only server functionality, no Specializations are described within this section.

3.3 Pre-Scenario communication

3.3.1 General information

The Pre-Scenario communication is needed if a client does not know the corresponding addresses on the server or if the required subscriptions or bindings are not active. In this case certain general communication mechanisms SHALL be used within SPINE:

- Detailed discovery: allows to discover resource addresses.
- Binding: allows to bind to resource address, which is frequently necessary to obtain write privileges.
- Subscription: allows to subscribe to resource addresses, which is necessary to receive unsolicited notifications if a resource changes during runtime.

It is possible to combine those steps for multiple Scenarios or also multiple Use Cases:

- E.g. if multiple Scenarios in multiple Use Cases use the same Feature, only one subscription needs to occur.
- E.g. a complete detailed discovery or a subscription to the NodeManagement Feature needs to occur only once for all Use Cases.

Depending on which Entity, Feature and Functions are used within a Scenario the payload of the corresponding messages may slightly differ, but the basic principles and messages used stay the same.

The subsequent messages SHALL be exchanged for those parts that have not already been performed since the current connection is established or if those parts are outdated for another reason (e.g. if the detailed discovery is needed, but the bindings and subscriptions are still active from a previous connection only the detailed discovery messages need to be exchanged). If all Pre-Scenario communication parts are up to date, this section MAY be skipped, and the implementation can proceed as described in the corresponding "Scenario communication" sections.

After the connection is re-established (e.g. due to a power loss or a firmware update) the Pre-Scenario communication SHALL be performed as well. There may be circumstances where messages from the Pre-Scenario communication may be exchanged again.

Often the necessary messages of different Scenarios can be combined, so that only one single message is needed instead of multiple messages for the different Scenarios. This also is the case for the Pre-Scenario communication. In most cases only one "read" operation on the detailed discovery is necessary, as well as only one subscription request or binding request is needed for each Feature. Often multiple Scenarios within a Use Case access the same Feature, so only one subscription or binding is necessary.

3.3.2 Detailed discovery

For the functionality where a client already has current detailed discovery information (i.e. independent of this Use Case or any Scenario of it) the remainder of this section SHOULD be skipped.

Otherwise, the following procedure SHALL be performed in the given order:

1. If a client is not subscribed to the primary NodeManagement instance, the client SHALL acquire a subscription according to the figure provided within this sub-section.
2. A client SHALL read the detailed discovery information according to the figure provided within this sub-section. It SHALL keep the received information as far as it concerns mandatory and supported optional Entity Types, Feature Types and Functions of this Use Case that are needed by the client. This means that a client may choose how to store the necessary information. E.g. a client Actor can store the information how to address the necessary Features of the implemented Scenarios but may discard the Entity information.
3. If and as long as a client has a subscription to the detailed discovery information of an Actor and receives proper notifications, it SHALL consider (i.e. integrate into the kept detailed discovery information) the received information as far as it concerns mandatory and supported optional Entity Types, Feature Types and Functions of this Use Case.

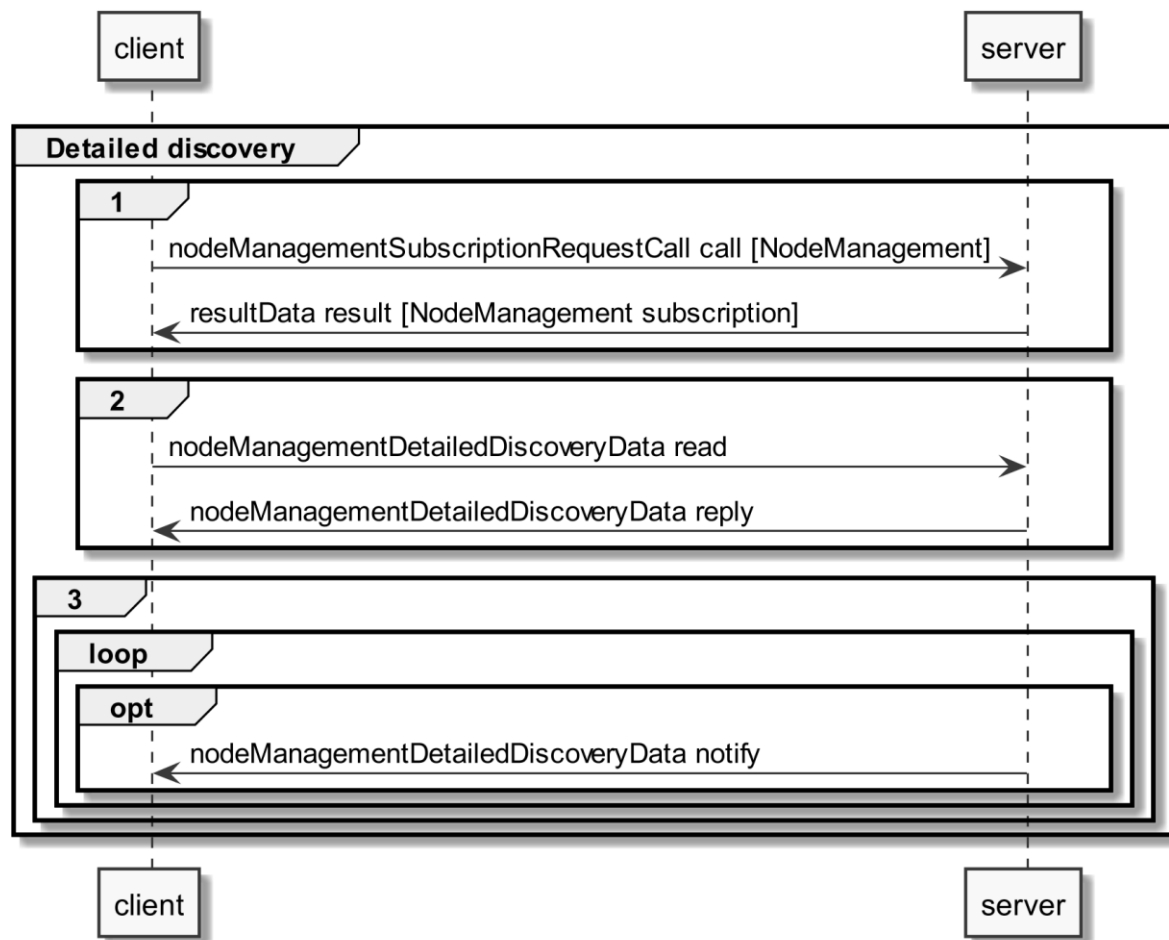


Figure 12: Pre-Scenario communication - Detailed discovery sequence diagram

If the "nodeManagementDetailedDiscoveryData read" fails, the client SHOULD retry to read the detailed discovery information until the "nodeManagementDetailedDiscoveryData reply" message was received successfully.

If all functionality is present at all times: The "nodeManagementDetailedDiscoveryData reply" message contains at least the mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as the used Functions and their "possible operations" described in section 3.2 and its sub-sections.

If functionality is added or removed dynamically: The "nodeManagementDetailedDiscoveryData reply" message does not need to contain all mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as all needed Functions and their "possible operations" described in section 3.2 and its sub-sections. However, as soon as the functionality is available it will be announced via a "nodeManagementDetailedDiscoveryData notify" message.

For the nodeManagementDetailedDiscoveryData read Function it is recommended to use a partial read with separated Selectors that may use one of the following Elements:

- entityType
- featureType

Note: Even with the usage of Selectors Features and Entities that are not relevant for this Use Case may be discovered. However, only Features and Entities that fulfil the hierarchical order as described within the Actors' sections shall be considered for this Use Case.

A "partial" notify SHALL be supported without using Selectors and Elements. Partial "delete" notify SHOULD also be supported with separated Selectors that may use one of the following Elements:

- entityAddress
- featureAddress

3.3.3 Binding

If binding is required by a Scenario that uses Features with writeable or changeable data, the server SHALL support binding for the respective Features. Before a write on a Function of a Feature occurs, the client SHALL create a binding to the corresponding Feature. For this the nodeManagementBindingRequestCall Function is used as shown in the following sequence diagram:

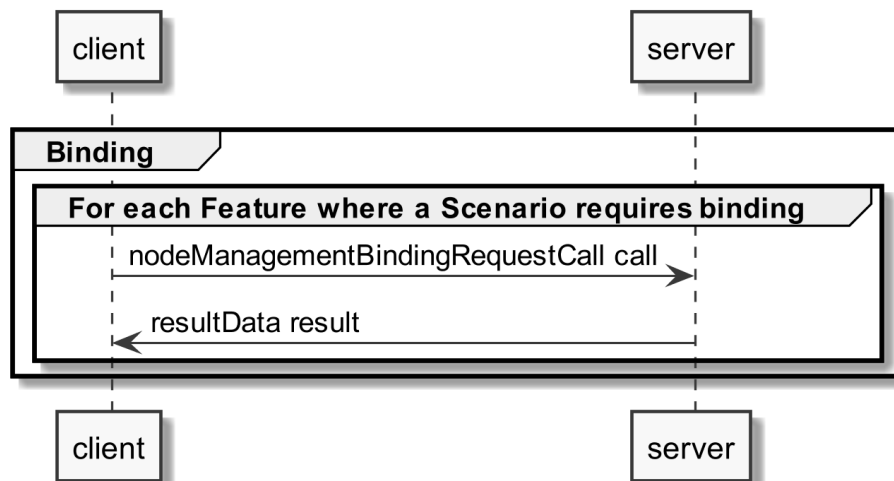


Figure 13: Pre-Scenario communication - Binding sequence diagram

If functionality is added or removed dynamically, binding may not be possible at all times on the required Functions. A client SHALL retry to create a binding again when receiving according updated detailed discovery information.

3.3.4 Subscription

A server SHALL support subscription for all Features that contain readable data that may change during runtime. The client SHALL create a subscription for all Features that the client wants to read. For this the nodeManagementSubscriptionRequestCall Function is used as shown in the following sequence diagram:

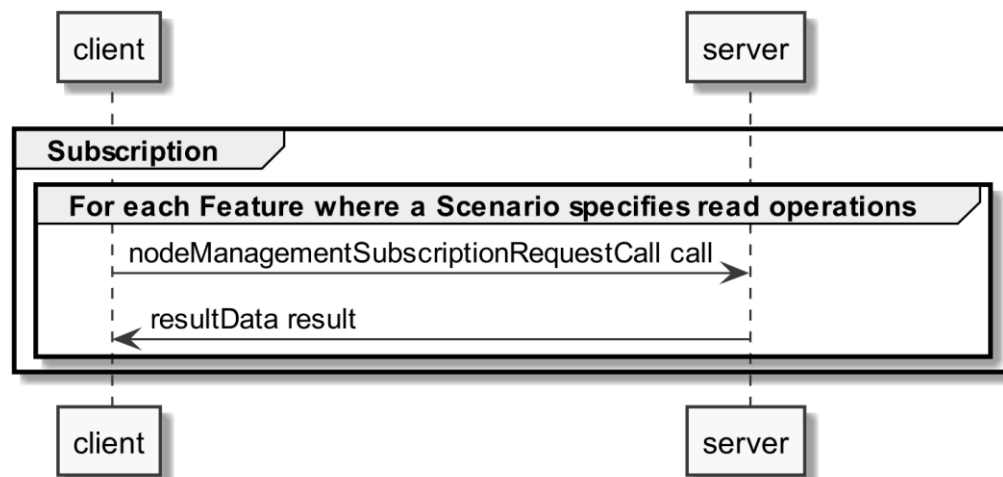


Figure 14: Pre-Scenario communication - Subscription sequence diagram

If the subscription request fails (e.g. because it is not supported by the server or the maximum number of possible subscriptions is reached), the client SHOULD read the data periodically (so-called "polling").

If functionality is added or removed dynamically, subscription may not be possible at all times on the required Functions. A client SHALL retry its subscription procedure again when receiving according updated detailed discovery information.

3.3.5 Dynamic behaviour

In case Entities or Features are removed, a nodeManagementDetailedDiscoveryData "notify" is transmitted that informs about the deleted Entities and Features. All existing binding or subscription entries on the deleted Features SHALL be deleted by each device.

In case Entities or Features are added the Pre-Scenario communication starts with transmitting a nodeManagementDetailedDiscoveryData "notify" that contains the added Entities and Features.

3.4 Scenarios

3.4.1 Scenario 1 - Monitor heat pump compressor's power consumption flexibility

3.4.1.1 Pre-Scenario communication

1. **Detailed discovery:** Actors that act as client within this Scenario need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding:** Binding SHOULD NOT be used for this Scenario. Please note, that Scenario 2 works with the same Feature used for Scenario 1 and has a different requirement for binding.
3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.1.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

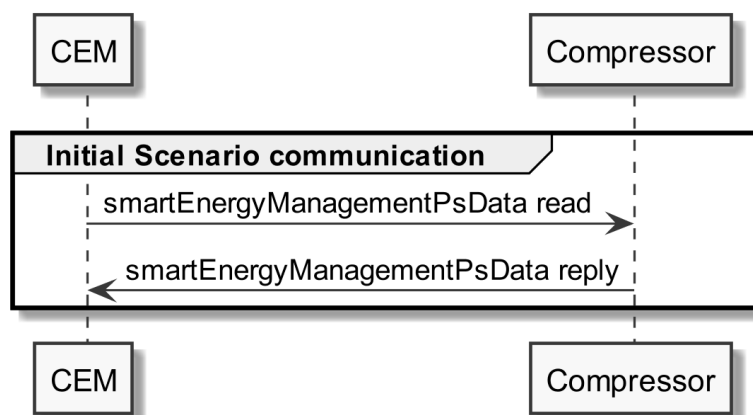


Figure 15: Scenario 1 - Initial Scenario communication sequence diagram

The following table shows where the required content of the messages from the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
smartEnergyManagementPsData reply	Table 11	1

Table 12: Initial Scenario communication content references for Scenario 1

Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be provided completely, but later during Runtime Scenario communication.

3.4.1.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

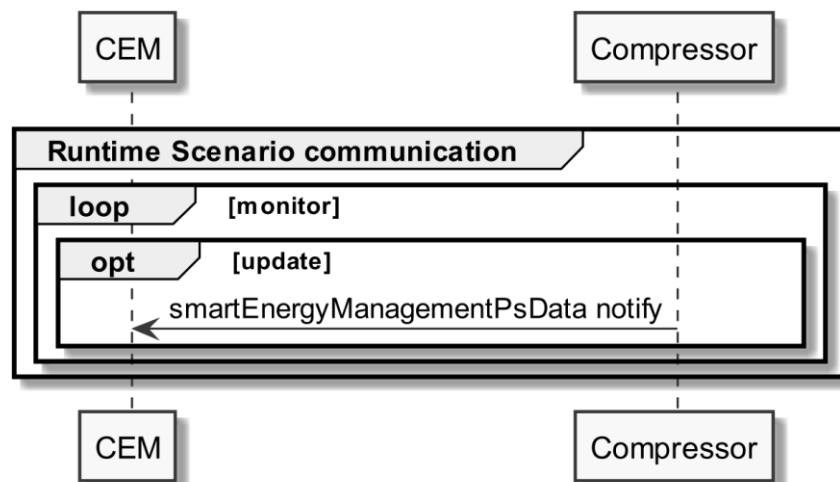


Figure 16: Scenario 1 - Runtime Scenario communication sequence diagram

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For smartEnergyManagementPsData "partial" delete notifications SHOULD be supported with the Selector:

- sequenceId

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
smartEnergyManagementPsData notify	Table 11	1

Table 13: Runtime Scenario communication content references for Scenario 1

3.4.1.4 Additional information

3.4.1.4.1 Additional rules for power sequences:

At least one of the two Elements SHALL be set to true [OHPCF-011/7]:

- alternatives. powerSequence. operatingConstraintsInterrupt. isPausable
- alternatives. powerSequence. operatingConstraintsInterrupt. isStoppable

3.4.2 Scenario 2 - Control heat pump compressor's power consumption flexibility

See section 3.4.1, as Scenario 1 is the basis for Scenario 2. In addition, the following applies:

1. **Binding:** Actors that write parts of a Feature within this Scenario need to create a binding as described in section 3.3.3. Only one binding partner is allowed to write the data specified in this Scenario.

3.4.2.1 Initial Scenario communication

See section 3.4.1, as Scenario 1 is the basis and potential trigger for Scenario 2.

3.4.2.2 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

The Actor CEM executes Scenario 2 by sending the required change to Actor Compressor as shown in the following figure:

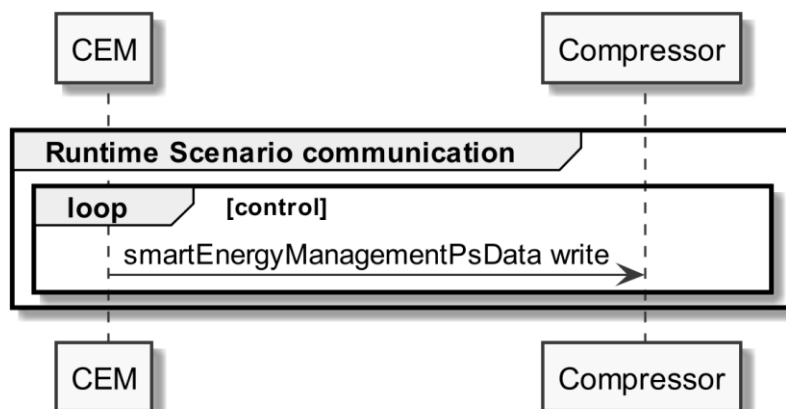


Figure 17: Scenario 2 - Runtime Scenario communication sequence diagram

For smartEnergyManagementPsData "partial" write operations SHALL be supported with the Selector:

- sequenceld

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
smartEnergyManagementPsData write	Table 11	2

1161 *Table 14: Runtime Scenario communication content references for Scenario 2*

1162

1163 **3.4.2.3 Additional information**

1164 None.

1165